

Exploring the dynamical interplay between mass-energy equivalence, interactions and entanglement in an optical lattice clock: Supplemental Materials

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(Dated: January 16, 2025)

S1. GENERAL RELATIVISTIC CORRECTIONS IN OPTICAL LATTICE CLOCKS

Our discussion is based on the post-Newtonian expansion of Schwarzschild metric [S1–S4] in the isotropic coordinates (ct, X, Y, Z) ,

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = -\left(1 + 2\frac{\phi}{c^2} + 2\frac{\phi^2}{c^4}\right)c^2 dt^2 + \left(1 - 2\frac{\phi}{c^2}\right)(dX^2 + dY^2 + dZ^2) + O\left(\frac{1}{c^4}\right), \quad (\text{S1})$$

where $\phi \approx g_{\text{LO}}Z$ is the Newtonian gravitational potential, with g_{LO} the local gravitational acceleration. This metric defines the lab frame as it reduces to Minkowski metric at the lab position $X = Y = Z = 0$. Here we list all the relevant GR correction terms for the single-atom Hamiltonian $\hat{H}_{\text{sp}}$ in the lab frame,

$$\hat{H}_{\text{sp}} = \hat{H}_A + \hat{H}_{AL}, \quad (\text{S2})$$

and one can refer to Refs. [S2, S3] for the details of the derivation. $\hat{H}_A$ is the atom Hamiltonian including center of mass and internal degrees of freedom,

$$\hat{H}_A = \hat{H}_{\text{point}}\left(\hat{\mathbf{X}}, \hat{\mathbf{P}}, M + \frac{\hat{H}_I}{c^2}\right) + O\left(\frac{1}{c^4}\right), \quad (\text{S3})$$

where $\hat{H}_I$ is the internal Hamiltonian, and M is the rest mass of a ground-state atom in center-of-mass coordinates. Here $\hat{H}_{\text{point}}$ is the Hamiltonian of a point particle,

$$\hat{H}_{\text{point}}(\hat{\mathbf{X}}, \hat{\mathbf{P}}, M) = Mc^2 + \frac{\hat{\mathbf{P}}^2}{2M} + M\phi + \hat{H}_{\text{other}}, \quad (\text{S4})$$

where $\hat{H}_{\text{other}}$ contains the relativistic corrections,

$$\hat{H}_{\text{other}} = -\frac{\hat{\mathbf{P}}^4}{8M^3c^2} + M\frac{\phi^2}{2c^2} + \frac{3}{2}\frac{\hat{\mathbf{P}} \cdot \phi \hat{\mathbf{P}}}{Mc^2} + O\left(\frac{1}{c^4}\right). \quad (\text{S5})$$

$\hat{H}_{AL}$ describes the couplings to external electromagnetic field. Although the higher order terms in multipolar expansion can play a more important role compared to the GR corrections, here we only focus on the GR corrections to the leading order electric dipole terms. Under electric dipole approximation, we have [S2, S5]

$$\hat{H}_{AL} = -\hat{\mathbf{d}} \cdot \mathbf{E} + \frac{1}{2M} \left[\hat{\mathbf{P}} \cdot (\hat{\mathbf{d}} \times \mathbf{B}) + (\hat{\mathbf{d}} \times \mathbf{B}) \cdot \hat{\mathbf{P}} \right], \quad (\text{S6})$$

where $\hat{\mathbf{d}}$ is the electric dipole moment. In $\hat{H}_{AL}$, the first term is the standard electric dipole coupling, and the second term is a relativistic correction term known as the Röntgen term, describing the coupling between a moving dipole and a magnetic field. The electromagnetic field in $\hat{H}_{AL}$ also contains GR corrections due to the metric in the lab frame, as pointed out in Ref. [S4]. For example, the electric field of a plane wave propagating along $\vec{Z}$ direction (electric field polarized along $\vec{X}$ direction) should be corrected in the following way,

$$\mathbf{E} \propto \left(\left(1 - 2\frac{\phi}{c^2}\right) \exp \left[-i\omega t \pm \left(1 - \frac{\phi}{c^2}\right) ikZ \right] + \text{c.c.}, 0, 0 \right), \quad (\text{S7})$$

where $k = \omega/c$. In the following, we would like to analyze the order of magnitude for all these GR corrections in optical lattice clocks.

A. Atomic center-of-mass coordinates

GR corrections in the atomic center-of-mass coordinates are given by

1. Mass-energy equivalence. As shown in Eq. (S3), the atom Hamiltonian $\hat{H}_A$ can be generated by replacing mass M by $M + \hat{H}_I/c^2$ in the point-particle Hamiltonian $\hat{H}_{\text{point}}$. In other words, the mass of a composite particle comprises the rest masses of the constituent particles as well as the internal energy. The replacement in the kinetic energy term $\hat{\mathbf{P}}^2/2M$ can lead to second-order Doppler shift (also known as motional time dilation in special relativity),

$$\hat{H}_{\text{SDS}} = -\frac{\hat{\mathbf{P}}^2}{2M} \frac{\hat{H}_I}{Mc^2}. \quad (\text{S8})$$

The replacement in the gravitational potential energy term $M\phi$ can lead to gravitational redshift (also known as gravitational time dilation),

$$\hat{H}_{\text{GRS}} = \frac{\phi}{c^2} \hat{H}_I. \quad (\text{S9})$$

We have discussed these two types of corrections in the main text, since they are the dominant GR corrections for optical lattice clocks.

2. $\hat{H}_{\text{other}}$ (see Eq. (S5)). Compared to mass-energy equivalence, $\hat{H}_{\text{other}}$ only acts on center-of-mass degrees of freedom and does not lead to frequency shifts of the clock transition. The observation of these corrections requires coherent superposition of wave packets at different positions, which is beyond the scope of our manuscript. For completeness, we still analyze their orders of magnitude in an optical lattice clock.

In a relatively deep optical lattice, to the leading order we can approximate each lattice sites as an harmonic oscillator with trapping frequencies $\omega_\alpha = 2\sqrt{V_\alpha E_R}/\hbar$, with $\alpha = X, Y, Z$ and E_R the recoil energy. In the ground band of the optical lattice, we have

$$\left\langle \frac{\hat{\mathbf{P}}^2}{2M} \right\rangle \approx \sum_\alpha \frac{\hbar\omega_\alpha}{4}, \quad \left\langle \frac{\hat{\mathbf{P}}^4}{8M^3c^2} \right\rangle \approx \sum_\alpha \frac{(\hbar\omega_\alpha)^2}{16Mc^2} + \sum_{\alpha\alpha'} \frac{(\hbar\omega_\alpha)(\hbar\omega_{\alpha'})}{32Mc^2}. \quad (\text{S10})$$

For $V_{X,Y,Z} = 300E_R$, we have $\langle \hat{\mathbf{P}}^4/8M^3c^2 \rangle \sim 7.9 \times 10^{-31}$ in fractional frequency unit. Since the Wannier-Stark states are localized within very few lattice sites, it is reasonable to consider Z as a c-number, leading to $\langle M\phi^2/2c^2 \rangle \sim 2.7 \times 10^{-26}$ with 1cm separation, and 2.7×10^{-22} with 1m separation. Similarly, we have $\langle \hat{\mathbf{P}} \cdot \phi \hat{\mathbf{P}}/Mc^2 \rangle \sim 3.0 \times 10^{-28}$ with 1cm separation, and 3.0×10^{-26} with 1m separation. These are tiny corrections compared to the mass-energy equivalence.

B. Atomic internal coordinates

GR corrections in the atomic internal coordinates ($\hat{H}_I$) are given by

1. Special relativistic corrections. These are indeed the leading order corrections of $\hat{H}_I$ based on the Dirac equation, but they are already included in the atomic physics calculations that give us information of the intrinsic atomic structure of the atom in consideration. For example, the ground and excited states we considered in the manuscript are the 1S_0 and 3P_0 states of a Sr atom, and these include corrections due to the spin-orbit coupling effect. Basically, they have been included already in the determination of the optical transition frequency ω_0 .
2. Gravitational corrections. These corrections can be estimated by extending the Dirac equation into curved spacetime, and one can refer to Refs. [S6–S10]. For a one-electron atom, the frequency shift due to gravity is typically at the order of $a_0 r_s/R^2$ with respect to the electron rest mass (see Ref. [S9]), where a_0 is the Bohr radius, r_s is the Schwarzschild radius of the earth, and R is the coordinate radius with respect to the gravitational center. If the same results apply to Sr atoms, then the frequency shift will be typically below 10^{-23} in fractional frequency units. These are tiny corrections compared to the mass-energy equivalence.

C. Optical lattice potential

GR corrections of the optical lattice potential are given by

1. Special relativistic correction (Röntgen term). We consider monochromatic light field in the following form, $\mathbf{E} = \mathbf{E}^{(+)}e^{-i\omega t} + \text{H.c.}$, $\mathbf{B} = \mathbf{B}^{(+)}e^{-i\omega t} + \text{H.c.}$, where ω is the laser frequency, and assume the lattice AC Stark shift is dominated by a single dipole-allowed transition $|a\rangle \rightarrow |b\rangle$ for simplicity, so the AC Stark shift for internal state $|a\rangle$ is given by second-order perturbation theory,

$$\hat{H}_{\text{ac}} \approx -\frac{\langle a|\hat{H}_{\text{AL}}|b\rangle\langle b|\hat{H}_{\text{AL}}|a\rangle}{E_b - E_a - \hbar\omega_L}, \quad (\text{S11})$$

where ω_L is the frequency of the lattice beam, E_a, E_b are the energy of states $|a\rangle$ and $|b\rangle$ respectively. Here we assume linear polarization of the lattice beam, with $\mathbf{E}^{(+)} = \mathcal{E} \cos(k_L Z) \vec{X}$, $\mathbf{B}^{(+)} = (i\mathcal{E}/c) \sin(k_L Z) \vec{Y}$ and propagation direction $\vec{Z}$. Here the vector symbol $\vec{X}$ means a unit vector along X direction. Plugging in Eq. (S11), we have

$$\hat{H}_{\text{ac}} \approx -\alpha_{\text{E1}} \mathcal{E}^2 \cos^2(k_L Z) + \alpha_{\text{E1}} \mathcal{E}^2 \frac{\hbar\omega_L}{Mc^2} \sin^2(k_L Z), \quad (\text{S12})$$

where $\alpha_{\text{E1}} = \langle a|\hat{\mathbf{d}} \cdot \vec{X}|b\rangle\langle b|\hat{\mathbf{d}} \cdot \vec{X}|a\rangle/(E_b - E_a - \hbar\omega)$ is the electric dipole polarizability. Here, the first term is the standard optical lattice potential, and the second term is the corrections due to the Röntgen term. This leads to a correction of the lattice depth at the order of $\hbar\omega_L/Mc^2 \sim 10^{-11}$, which is equivalent to a fractional frequency of 10^{-20} for a MHz trapping potential. However, this term does not lead to frequency shifts of the clock transition in a magic wavelength lattice, which shares the same α_{E1} for the ground and excited states of the clock transition.

2. Gravitational correction to electromagnetic waves. As shown in Eq. (S7), we have ϕ/c^2 corrections in the amplitude and phase of the electric field. Note that this term does not lead to differential AC Stark shifts, while it leads to a spatially dependent lattice depth and lattice spacing. Considering a MHz trapping potential, such type of correction is at the order of 10^{-27} for 1cm separation and 10^{-25} for 1m separation in fractional frequency units.

Therefore, we can ignore the GR corrections of the optical lattice potential to the leading order in our case.

D. Photon-mediated interactions

Photon-mediated interactions (e.g. collective spin exchange interaction $\hat{S}^+ \hat{S}^-$) originate from adiabatic elimination of cavity photons in the Tavis-Cumming Hamiltonian,

$$\hat{H}_{\text{TC}}/\hbar = -\Delta_c \hat{a}^\dagger \hat{a} + g_c (\hat{a} \hat{S}^+ + \hat{a}^\dagger \hat{S}^-), \quad (\text{S13})$$

where $\hat{a}$ is annihilation operator of the cavity mode, $\Delta_c = \omega_0 - \omega_c$ is the detuning between atomic transition frequency ω_0 and cavity resonance ω_c , and $2g_c$ is the single-photon Rabi frequency describing the strength of atom-cavity coupling. For simplicity, we assume homogeneous atom-cavity coupling among all the atoms. As for GR corrections, it would be simpler to discuss them in the Tavis-Cumming Hamiltonian $\hat{H}_{\text{TC}}$.

1. Special relativistic correction (Röntgen term). Plug in $\hat{H}_{\text{AL}}$ the quantized electromagnetic fields of a standing-wave cavity mode, $\hat{\mathbf{E}}^{(+)} = \sqrt{\hbar\omega_c/2\epsilon_0 V} \cos(k_c Z) \hat{a} \vec{X}$, $\hat{\mathbf{B}}^{(+)} = (i/c) \sqrt{\hbar\omega_c/2\epsilon_0 V} \sin(k_c Z) \hat{a} \vec{Y}$, where V is the mode volume. The atom-cavity coupling is given by

$$g_c = -\sqrt{\frac{\hbar\omega_c}{2\epsilon_0 V}} \frac{\langle e|\hat{\mathbf{d}} \cdot \vec{X}|g\rangle}{\hbar} \left[\int d^3\mathbf{R} \psi_e^* \cos(k_c Z) \psi_g - \frac{i}{2Mc} \int d^3\mathbf{R} \psi_e^* \left(\hat{P}_Z \sin(k_c Z) + \sin(k_c Z) \hat{P}_Z \right) \psi_g \right]. \quad (\text{S14})$$

Therefore, GR corrections of g_c are at the order of $\langle \hat{P}_Z \rangle / Mc \sim 10^{-11}$. If we consider the cavity mode is coupled to the carrier transition of the atoms without changing the motional degrees of freedom (motional wave function $\psi_e \approx \psi_g$), GR corrections would be further suppressed since $\hat{P}_Z \sin(k_c Z) + \sin(k_c Z) \hat{P}_Z$ is off-diagonal under a harmonic approximation of the optical lattice. Considering photon-mediated interaction at the order of Hz [S11], GR corrections due to Röntgen term would be 10^{-26} or smaller in fractional frequency units.

2. Gravitational correction to cavity mode function. As shown in Eq. (S7), we have ϕ/c^2 corrections in the cavity mode function, leading to corrections of the overlap integral in Eq. (S14) as well as the cavity resonance ω_c due to the change of round-trip phase. However, similar to the discussion of the optical lattice potential, the ϕ/c^2 corrections in cavity-mediated interactions are negligible due to the small spatial separation of the optical clocks. For 1m spatial separation, we have $\phi/c^2 \sim 10^{-16}$, leading to corrections in the atom-cavity coupling below 10^{-31} in fractional frequency units.

Therefore, we can ignore the GR corrections of the photon-mediated interactions to the leading order in our case.

S2. DETAILED ANALYSIS OF EMERGENT SYNCHRONIZATION

In the main text, we discuss the interplay between atomic interactions, entanglement and gravitational redshift in Fig. 3 and Fig. 4. Here we would like to provide analytic spin-wave calculations based on two different approaches, including Holstein–Primakoff approximation and restriction of dynamics within collective and spin-wave manifold, and finally discuss the experimental considerations of our protocol.

First we would like to provide numerical results for the validity of the spin-wave calculation. As shown in Fig. S1(a,b), we calculate the expectation value $\langle \hat{\mathbf{S}} \cdot \hat{\mathbf{S}} \rangle$ as a function of the evolution time, using the same time sequence as Fig. 3 in the main text. Note that in a given total spin- S manifold, we have $\langle \hat{\mathbf{S}} \cdot \hat{\mathbf{S}} \rangle = S(S+1)$. So $\langle \hat{\mathbf{S}} \cdot \hat{\mathbf{S}} \rangle$ can serve as a measure of population in the collective manifold ($S = N/2$), spin-wave manifold ($S = N/2 - 1$), as well as the manifolds with lower total spin. We consider cavity-mediated interaction $\hat{H}_{\text{cav}}$ with $J_{\perp} = J_z$ (see Eq. (5) in the main text). In Fig. S1(a), we set $\omega_{\text{split}}/NJ_{\perp} = 0.3125$, which is the same parameter for global frequency synchronization shown in Fig. 3(b,d) in the main text. In this case we find undamped oscillations between the collective manifold ($S = N/2$) and the spin-wave manifold ($S = N/2 - 1$), ensuring the validity of the spin-wave calculation. In Fig. S1(b), we set $\omega_{\text{split}}/NJ_{\perp} = 3.125$, which is the same parameter for Fig. 3(c,e) in the main text, where global frequency synchronization fails to occur. In this case the system can evolve to manifolds with lower total spin, so the spin-wave calculation no longer valid. The fast growth of entanglement shown in Fig. 3(e) in the main text might due to the population in the highly entangled states in manifolds with lower total spin.

In the following, we focus on the regime of global frequency synchronization. We define $\hat{\mathcal{S}}_j^{x,y,z}$ as collective spin operators summed over all atoms at the same height ja_L , and consider N_s lattice sites in Z direction. Based on our protocol in Fig. 3 and Fig. 4 in the main text, the frequency of each atom $\omega_j(t)$ can be estimated by

$$\omega_j(t) = \frac{1}{t} \arctan \left(\frac{\langle \hat{\mathcal{S}}_j^y(t) \rangle}{\langle \hat{\mathcal{S}}_j^x(t) \rangle} \right). \quad (\text{S15})$$

The rigorous definition of the synchronization time t_{syn} is the time for the first minimum in the variance of atomic frequencies. In fact, we can approximately reach the zero crossing of $\omega_j(t)$ for all the j at t_{syn} , as demonstrated by the analytic results below and by numerical results shown in Fig. 3(b) in the main text.

A. Holstein–Primakoff approximation

We would like to use the Holstein–Primakoff approximation to describe the case with an unentangled initial state (see Fig. 3 in the main text). We consider $N = N_{\perp} N_s$ spins with all the spins pointing to $+x$ direction initially, where $S = N_{\perp}/2$ is the total spin at the same height. Based on the Holstein–Primakoff approximation, we have

$$\hat{\mathcal{S}}_j^x = \frac{N_{\perp}}{2} - \hat{a}_j^{\dagger} \hat{a}_j, \quad \hat{\mathcal{S}}_j^y \approx \sqrt{N_{\perp}} \frac{\hat{a}_j + \hat{a}_j^{\dagger}}{2}, \quad \hat{\mathcal{S}}_j^z \approx \sqrt{N_{\perp}} \frac{\hat{a}_j - \hat{a}_j^{\dagger}}{2i}. \quad (\text{S16})$$

In this way, the initial state becomes the vacuum state of all these bosonic operators. In the following, we will plug these bosonic operators into the Hamiltonian and keep the terms up to quadratic order of bosonic operators. We then apply Fourier transform to obtain the bosonic operators for spin waves $k = 2\pi m/N_s$ with $m = 0, 1, 2, \dots, N_s - 1$,

$$\hat{a}_j = \frac{1}{\sqrt{N_s}} \sum_k e^{ikj} \hat{a}_k, \quad \hat{a}_j^{\dagger} = \frac{1}{\sqrt{N_s}} \sum_k e^{-ikj} \hat{a}_k^{\dagger}, \quad (\text{S17})$$

and rewrite the Hamiltonian accordingly. The validity of Holstein–Primakoff approximation requires $\langle \hat{a}_j \rangle \ll 1$ for all j . If we define $\eta = \omega_{\text{split}}/(NJ_{\perp})$ as the ratio between the maximum redshift in the array ($\omega_{\text{split}} = (N_s - 1)\omega_{\text{GRS}}$) and

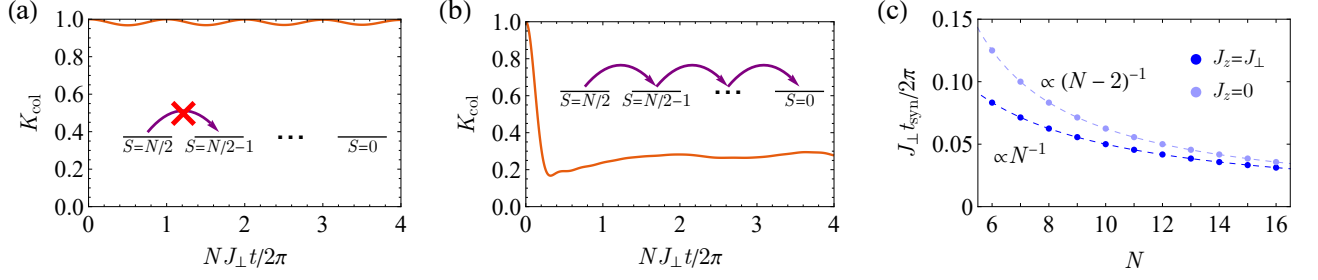


Fig. S1. (a) We evolve the system based on the time sequence in Fig. 3 in the main text and calculate the expectation value $K_{\text{col}} = \langle \hat{\mathbf{S}} \cdot \hat{\mathbf{S}} \rangle / [\frac{N}{2}(\frac{N}{2} + 1)]$ as a function of evolution time. We consider Hamiltonian $\hat{H}_{\text{cGR}}$ with $J_{\perp} = J_z$ (Eq. (5) in the main text). The choice of parameters is the same as Fig. 3(b,d) in the main text, $\omega_{\text{split}}/NJ_{\perp} = 0.3125$. (b) The same calculation as (a) with $\omega_{\text{split}}/NJ_{\perp} = 3.125$, which is the same as Fig. 3(c,e) in the main text. (c) Scaling of synchronization time t_{syn} as a function of atom number N for cavity-mediated interactions. We compare the t_{syn} between collective Heisenberg interactions ($J_z = J_{\perp}$) and spin exchange interactions ($J_z = 0$). All the calculations are based on an initial unentangled state with all the spins pointing to $+x$ direction.

the spin wave excitation gap ($NJ_{\perp}$ as discussed below), the typical condition for validity would be $\eta \ll 1$. In this regime, we have $\langle \hat{S}_j^x \rangle \approx N_{\perp}/2$, so the frequency of each atom $\omega_j(t)$ can be approximated as

$$\omega_j(t) \approx \frac{2\langle \hat{S}_j^y(t) \rangle}{N_{\perp}t}. \quad (\text{S18})$$

Here we consider the following Hamiltonian describing the interplay between collective Heisenberg interactions and gravitational redshift (see Eq. (5) in the main text),

$$\hat{H}_{\text{cGR}}/\hbar = J_{\perp} \sum_{i=0}^{N_s-1} \sum_{j=0}^{N_s-1} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \omega_{\text{GRS}} \sum_{j=0}^{N_s-1} \left(j - \frac{N_s-1}{2} \right) \hat{S}_j^z. \quad (\text{S19})$$

Applying the Holstein–Primakoff bosons and keeping the terms up to quadratic order, the Hamiltonian in terms of spin-wave operators becomes

$$\hat{H}_{\text{cGR}}/\hbar \approx -NJ_{\perp} \sum_{k \neq 0} \hat{a}_k^{\dagger} \hat{a}_k + \frac{\omega_{\text{GRS}}}{2i} \sqrt{N} \sum_{k \neq 0} \left(\frac{1}{e^{ik} - 1} \hat{a}_k - \frac{1}{e^{-ik} - 1} \hat{a}_k^{\dagger} \right). \quad (\text{S20})$$

Solving the Heisenberg equation of motion for $\hat{a}_k$ with $k \neq 0$, one can finally reach

$$\omega_j(t) \approx -\frac{\omega_{\text{GRS}}}{2} \frac{\sin(NJ_{\perp}t)}{NJ_{\perp}t} \sum_{k \neq 0} \frac{\sin(kj + k/2)}{\sin(k/2)} = \omega_{\text{GRS}} \frac{\sin(NJ_{\perp}t)}{NJ_{\perp}t} \left(j - \frac{N_s-1}{2} \right). \quad (\text{S21})$$

Similarly, $\omega_j(0)$ agrees with the gravitational redshift value without interactions. As for the frequency synchronization, it happens at $NJ_{\perp}t = \pi$, which gives

$$J_{\perp}t_{\text{syn}} = \frac{\pi}{N}. \quad (\text{S22})$$

This analytic result agrees with the numerical simulations (see Fig. S1(b)). Within the Holstein–Primakoff approximation, there is no deviation of $\omega_j(t_{\text{syn}})$ from 0.

B. Restriction within collective and spin-wave manifold for two large spins

For the case with an entangled initial state, it is not possible to use the Holstein–Primakoff approximation. Alternatively, we would like to simplify the Hamiltonian into two large spins ($S_1 = S_2 = N/4$) with effective Heisenberg interaction strength J_{eff} and effective redshift value ω_{eff} ,

$$\hat{H}_{\text{eff}}/\hbar = 2J_{\text{eff}} \hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 + \frac{\omega_{\text{eff}}}{2} (\hat{S}_1^z - \hat{S}_2^z). \quad (\text{S23})$$

In the regime $\omega_{\text{eff}} \ll NJ_{\text{eff}}$, we can restrict the dynamics within the collective manifold with total spin $S = N/2$ as well as the spin-wave manifold with total spin $S = N/2 - 1$. Based on the Clebsch-Gordan coefficients, the states in the collective manifold ($S = N/2$) can be expressed as

$$|N/2, m\rangle = \sum_{m_1} \sqrt{\frac{\binom{N/2}{N/4+m_1} \binom{N/2}{N/4+m-m_1}}{\binom{N}{N/2+m}}} |N/4, m_1\rangle_1 |N/4, m-m_1\rangle_2, \quad (\text{S24})$$

and the states in the spin-wave manifold ($S = N/2 - 1$) can be expressed as

$$|N/2 - 1, m\rangle = \sum_{m_1} (2m_1 - m) \sqrt{\frac{\binom{N/2}{N/4+m_1} \binom{N/2}{N/4+m-m_1}}{N \binom{N-2}{N/2-1+m}}} |N/4, m_1\rangle_1 |N/4, m-m_1\rangle_2, \quad (\text{S25})$$

where $\binom{n}{k}$ are binomial coefficients. A key observation from Eq. (S24) and Eq. (S25) is that

$$(\hat{S}_1^z - \hat{S}_2^z)|N/2, m\rangle = \sqrt{\frac{(N/2+m)(N/2-m)}{N-1}} |N/2-1, m\rangle. \quad (\text{S26})$$

If we restrict the dynamics within the collective and spin-wave manifold, $\hat{H}_{\text{eff}}$ can be reduced to 2×2 matrices in each m sector in the basis $\{|N/2, m\rangle, |N/2-1, m\rangle\}$,

$$\hat{H}_{\text{eff},m}/\hbar = \begin{pmatrix} NJ_{\text{eff}} & \frac{\omega_{\text{eff}}}{2} \sqrt{\frac{(N/2+m)(N/2-m)}{N-1}} \\ \frac{\omega_{\text{eff}}}{2} \sqrt{\frac{(N/2+m)(N/2-m)}{N-1}} & 0 \end{pmatrix}. \quad (\text{S27})$$

Also based on Eq. (S24) and Eq. (S25), we have ($j = 1, 2$)

$$\hat{S}_j^+ |N/2, m\rangle = \frac{1}{2} \sqrt{(N/2+m+1)(N/2-m)} |N/2, m+1\rangle - \frac{(-1)^j}{2} \sqrt{\frac{(N/2-m)(N/2-m-1)}{N-1}} |N/2-1, m+1\rangle, \quad (\text{S28})$$

$$\hat{S}_j^- |N/2, m\rangle = \frac{1}{2} \sqrt{(N/2-m+1)(N/2+m)} |N/2, m-1\rangle + \frac{(-1)^j}{2} \sqrt{\frac{(N/2+m)(N/2+m-1)}{N-1}} |N/2-1, m-1\rangle. \quad (\text{S29})$$

Now we consider a general initial state in the collective manifold,

$$|\psi_0\rangle = \sum_m c_m |N/2, m\rangle, \quad (\text{S30})$$

with the constraints $\langle \psi_0 | \hat{S}_n^y | \psi_0 \rangle = 0$, $\langle \psi_0 | \hat{S}_n^z | \psi_0 \rangle = 0$. With time evolution under $\hat{H}_{\text{eff}}$, we have

$$|\psi(t)\rangle = e^{-i\hat{H}_{\text{eff}}t/\hbar} \approx \sum_m c_m \left[e^{-iNJ_{\text{eff}}t/2} |N/2, m\rangle - i \frac{\omega_{\text{eff}}}{NJ_{\text{eff}}} \sqrt{\frac{(N/2)^2 - m^2}{N-1}} \sin(NJ_{\text{eff}}t/2) |N/2-1, m\rangle \right], \quad (\text{S31})$$

which gives

$$\begin{aligned} \langle \psi(t) | \hat{S}_j^+ | \psi(t) \rangle &\approx \sum_m c_m c_{m+1}^* \left[\langle N/2, m+1 | \hat{S}_j^+ | N/2, m \rangle \right. \\ &\quad - i \frac{\omega_{\text{eff}}}{NJ_{\text{eff}}} e^{iNJ_{\text{eff}}t/2} \sqrt{\frac{(N/2)^2 - m^2}{N-1}} \sin(NJ_{\text{eff}}t/2) \langle N/2, m+1 | \hat{S}_j^+ | N/2-1, m \rangle \\ &\quad \left. + i \frac{\omega_{\text{eff}}}{NJ_{\text{eff}}} e^{-iNJ_{\text{eff}}t/2} \sqrt{\frac{(N/2)^2 - (m+1)^2}{N-1}} \sin(NJ_{\text{eff}}t/2) \langle N/2-1, m+1 | \hat{S}_j^+ | N/2, m \rangle \right] \\ &= \sum_m c_m c_{m+1}^* \langle N/2, m+1 | \hat{S}_j^+ | N/2, m \rangle \left[1 - (-1)^j \frac{\omega_{\text{eff}}}{NJ_{\text{eff}}} \left(i \frac{1}{2} \sin(NJ_{\text{eff}}t) - \frac{2m+1}{N-1} \sin^2(NJ_{\text{eff}}t/2) \right) \right] \\ &= \langle \psi_0 | \hat{S}_j^+ | \psi_0 \rangle \left[1 - i(-1)^j \frac{\omega_{\text{eff}}}{2} \frac{\sin(NJ_{\text{eff}}t)}{NJ_{\text{eff}}} \right] + \frac{\langle \psi_0 | (\hat{S}_n^+ \hat{S}_n^z + \hat{S}_n^z \hat{S}_n^+) | \psi_0 \rangle}{N-1} (-1)^j \frac{\omega_{\text{eff}}}{NJ_{\text{eff}}} \sin^2(NJ_{\text{eff}}t/2). \end{aligned} \quad (\text{S32})$$

Therefore, we can obtain the frequency for each of the two large spins,

$$\omega_j(t) \approx (-1)^{j-1} \frac{\omega_{\text{eff}}}{2} \left(\frac{\sin(NJ_{\text{eff}}t)}{NJ_{\text{eff}}t} - 2 \frac{\langle \psi_0 | (\hat{S}^y \hat{S}^z + \hat{S}^z \hat{S}^y) | \psi_0 \rangle}{(N-1) \langle \psi_0 | \hat{S}^x | \psi_0 \rangle} \frac{\sin^2(NJ_{\text{eff}}t/2)}{NJ_{\text{eff}}t} \right), \quad (\text{S33})$$

leading to the synchronization time

$$NJ_{\text{eff}}t_{\text{syn}} = \pi - 2 \arctan \left[\frac{\langle \psi_0 | (\hat{S}^y \hat{S}^z + \hat{S}^z \hat{S}^y) | \psi_0 \rangle}{(N-1) \langle \psi_0 | \hat{S}^x | \psi_0 \rangle} \right]. \quad (\text{S34})$$

In the following, we consider three different types of initial states:

- $|\psi_0\rangle = |+\rangle^{\otimes N}$

In this case we have $\langle \psi_0 | (\hat{S}^y \hat{S}^z + \hat{S}^z \hat{S}^y) | \psi_0 \rangle = 0$, which gives

$$NJ_{\text{eff}}t_{\text{syn},0} = \pi. \quad (\text{S35})$$

Here we use $t_{\text{syn},0}$ to label the synchronization time with this unentangled initial state. This result agrees with the prediction using the Holstein–Primakoff approximation in the previous subsection.

- $|\psi_0\rangle = e^{-iQ\hat{S}^z/N} |+\rangle^{\otimes N}$

In this case we have

$$\langle \psi_0 | (\hat{S}^y \hat{S}^z + \hat{S}^z \hat{S}^y) | \psi_0 \rangle = \frac{N(N-1)}{2} \sin(Q/N) \cos^{N-2}(Q/N), \quad (\text{S36})$$

$$\langle \psi_0 | \hat{S}^x | \psi_0 \rangle = \frac{N}{2} \cos^{N-1}(Q/N), \quad (\text{S37})$$

which gives

$$NJ_{\text{eff}}t_{\text{syn}} = \pi - 2Q/N. \quad (\text{S38})$$

Note that the $\hat{S}^y \hat{S}^z$ term commutes with $\hat{H}_{\text{eff}}$, this result can also apply to the following Hamiltonian generated by cavity-mediated interactions,

$$\hat{H}_{\text{cav}}/\hbar = J_{\perp} \hat{\mathbf{S}} \cdot \hat{\mathbf{S}} + (J_z - J_{\perp}) \hat{S}^z \hat{S}^z + \frac{\omega_{\text{eff}}}{2} (\hat{S}_1^z - \hat{S}_2^z), \quad (\text{S39})$$

with initial state $|\psi_0\rangle = |+\rangle^{\otimes N}$. Now we can replace J_{eff} by $J_{\perp}$, and Q/N by $(J_z - J_{\perp})t_{\text{syn}}$, which gives

$$\left((N-2)J_{\perp} + 2J_z \right) t_{\text{syn}} = \pi. \quad (\text{S40})$$

We have shown numerical calculations for collective spin exchange interactions ($J_z = 0$) and collective Heisenberg interactions ($J_z = J_{\perp}$) with N atoms in Fig. S1(c), which agrees well with the analytic results using two large spins.

- $|\psi_0\rangle = e^{-i\theta\hat{S}^x} e^{-iQ\hat{S}^z/N} |+\rangle^{\otimes N}$

In this case we have

$$\langle \psi_0 | (\hat{S}^y \hat{S}^z + \hat{S}^z \hat{S}^y) | \psi_0 \rangle = \cos(2\theta) \frac{N(N-1)}{2} \sin(Q/N) \cos^{N-2}(Q/N) + \sin(2\theta) \frac{N(N-1)}{8} \left(1 - \cos^{N-2}(2Q/N) \right), \quad (\text{S41})$$

and $\langle \psi_0 | \hat{S}^x | \psi_0 \rangle$ is still given by Eq. (S37). So the tunability of synchronization time is due to the θ -rotation of $\langle \psi_0 | (\hat{S}^y \hat{S}^z + \hat{S}^z \hat{S}^y) | \psi_0 \rangle$. In Fig. 4 of the main text, we find that the analytic results for two large spins can also explain the numerical simulations for collective or nearest-neighbor Heisenberg interactions, if we rewrite the formula of synchronization time t_{syn} and its tunable range Δt_{syn} into dimensionless form,

$$\frac{t_{\text{syn}}}{t_{\text{syn},0}} = 1 - \frac{2}{\pi} \arctan \left[\frac{\langle \psi_0 | (\hat{S}^y \hat{S}^z + \hat{S}^z \hat{S}^y) | \psi_0 \rangle}{(N-1) \langle \psi_0 | \hat{S}^x | \psi_0 \rangle} \right], \quad (\text{S42})$$

$$\frac{\Delta t_{\text{syn}}}{t_{\text{syn},0}} = \frac{4}{\pi} \arctan \left[\frac{\langle \langle \psi_0 | (\hat{S}^y \hat{S}^z + \hat{S}^z \hat{S}^y) | \psi_0 \rangle_{\text{max}} \rangle}{(N-1) \langle \psi_0 | \hat{S}^x | \psi_0 \rangle} \right]. \quad (\text{S43})$$

In the main text, we use $|\psi(\theta)\rangle$ instead of $|\psi_0\rangle$ in this case to emphasize the θ -dependence.

C. Experimental considerations

Our discussions in the main text assume the frequency synchronization dynamics is purely unitary. Considering the existence of dissipative effects such as cavity photon loss, we would like to perform an order of magnitude analysis for the dissipative effects.

For simplicity, we analyze the case without the dressing scheme, so the $|\downarrow\rangle$ state is defined in the 1S_0 ground manifold of Sr, the $|\uparrow\rangle$ state is defined in the 3P_0 excited manifold, connected via an ultranarrow optical transition. This setup is the same as Ref. [S11], so the dynamics without dressing is captured by the following Lindblad master equation,

$$\frac{d}{dt}\hat{\rho} = -i[J_{\perp}\hat{S}^+\hat{S}^- + \hat{H}_{\text{GRS}}, \hat{\rho}] + \Gamma\left(\hat{S}^-\hat{\rho}\hat{S}^+ - \frac{1}{2}(\hat{S}^+\hat{S}^-\hat{\rho} + \hat{\rho}\hat{S}^+\hat{S}^-)\right), \quad (\text{S44})$$

where $J_{\perp} = g_c^2\Delta_c/(\Delta_c^2 + \kappa^2/4)$ is the spin exchange interaction strength, and $\Gamma = g_c^2\kappa/(\Delta_c^2 + \kappa^2/4)$ is the superradiant decay rate (Lindblad jump operator $\hat{S}^-$ originated from cavity loss). Here $2g_c$ is the single-photon Rabi frequency, Δ_c is the cavity detuning with respect to the atomic transition, and κ is the cavity linewidth. The ratio $\Gamma/J_{\perp}$ is determined by

$$\frac{\Gamma}{J_{\perp}} = \frac{\kappa}{\Delta_c}. \quad (\text{S45})$$

In the regime where $\Delta_c \gg \kappa$, the dynamics can be considered as near-unitary dynamics in the time scale $N\Gamma t \ll 1$.

If we use the same cavity as in Ref. [S11], with $g_c/2\pi = 4\text{Hz}$, and $\kappa/2\pi = 160\text{kHz}$, we can set $\Delta_c/2\pi = 5\text{MHz}$ with $N = 10^5$ atoms, leading to $NJ_{\perp}/2\pi = 0.32\text{Hz}$, $N\Gamma/2\pi = 0.01\text{Hz}$. Other dissipative effects such as spontaneous emission occurs at a rate $\gamma/2\pi \sim 1\text{mHz}$, which is smaller compared to superradiant decay. Including the dressing scheme, we will have an additional dissipative effect known as collective dephasing (Lindblad jump operator $\hat{S}^z$), which occurs at a comparable rate as the superradiant decay.

The frequency synchronization occurs at a time scale $NJ_{\perp}t_{\text{syn}}/2\pi \sim 0.5 \Rightarrow t_{\text{syn}} \sim 1.6\text{s}$, and one can check that $N\Gamma t_{\text{syn}} \sim 0.1 \ll 1$. So it is possible to achieve frequency synchronization within 2s, while in this time scale the dissipative effects remain subdominant. Dissipative effects would be more significant for highly entangled states. At the synchronization time scale, a lower bound of spin squeezing can be estimated by $\min(\xi^2) \sim N\Gamma t_{\text{syn}} \sim 0.1$. Therefore, the system dynamics is still approximately unitary for initial spin squeezed states below 10dB. Note that all these estimations are based on experimental parameters in Ref. [S11], and there is no fundamental limit in the experiment to reduce the cavity linewidth κ and further suppress the cavity induced dissipative effects.

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